

SAWERA NADEEM

MS Computer Science • Web Development Educator & Trainer • MERN Stack Developer

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PERSONAL STATEMENT

Web Development Educator and MS Computer Science student with a strong technical foundation in the MERN stack and Laravel, combined with extensive teaching experience across government, private, and university-affiliated platforms. Experienced in curriculum design, mentorship, and public engagement through invited seminars and guest lectures at multiple universities. Motivated to bring this dual technical-and-teaching background into academia — contributing to student learning and curriculum development at the university level while continuing to teach applied coding and web development courses.

EDUCATION

MS Computer Science

2025

BS Software Engineering — CGPA 3.7 | University of Agriculture, Faisalabad (UAF)

Key Courses: Web Development, Machine Learning, WordPress, Data Structures & Algorithms

TEACHING & ACADEMIC EXPERIENCE

Web Development & Coding Instructor | Saylani Mass IT Training Center

Selected to teach web development and coding at Saylani Mass IT Training Center, one of Pakistan's largest free technical-training networks.

Web Development & Coding Instructor | Virtual University of Pakistan — Government-Affiliated Course

Delivered on-site instruction in web development and coding as part of a government-affiliated course held at Virtual University.

Government Trainer | NAVTTC Program

Conducting a 3-month Laravel Full-Stack course covering MERN/MEAN.

Technologies: HTML, CSS, Bootstrap, JavaScript, PHP, jQuery, Git/GitHub, MySQL, Laravel

Instructor & Mentor | Technolangs Solutions

2024

- Delivered full-stack training in HTML, CSS, JavaScript, Bootstrap, MySQL, and PHP
- Conducted quizzes, hands-on labs, and real-world projects

Web Development Trainer (Remote) | 5 Batches Delivered

Trained national and international students through live coding, structured curriculum, and project-based learning.

Web Development Team Head | Senior Tutor Office — UAF

- Managed and guided a team of university-level developers working on academic and administrative software solutions
- Oversaw project structure, version control, and code quality

GUEST LECTURES, SEMINARS & INVITED TALKS

Invited Trainer | University of Central Punjab (UCP)

Delivered seminars and hands-on workshops on modern web development and career paths. Honoured with Guest Speaker Shield.

Invited Trainer | University of Agriculture, Faisalabad (UAF)

Conducted workshops on MERN stack fundamentals, freelancing, and practical development. Honoured with Guest Speaker Shield.

Guest Speaker | National Incubation Center (NIC), Faisalabad

Session on web development, startup tech skills, and the freelancing roadmap. Honoured with Guest Speaker Shield.

Speaker — Big Data Analytics Seminar | In collaboration with Sir Hafiz Ali

Delivered a seminar on Big Data Analytics in collaboration with Sir Hafiz Ali.

Moderator — Python for Beginners | iCodeGuru

2024

Managed live sessions, Q&A, and guided learners through Python basics.

Instructor | Freelancing Club (UAF Collaboration)

Teaching HTML, CSS, JavaScript, and professional project development.

INDUSTRY & PROFESSIONAL EXPERIENCE

Team Lead & Marketing Head | *Techloset Solutions*

Worked directly with Techloset CEO Naveed Sawar, leading the development team while heading marketing strategy and content. Naveed Sawar has also delivered an invited talk at University of Central Punjab (UCP).

Full Stack Developer | *TechHub — Faisalabad*

2023 – 2024

- Developed scalable MERN applications and created RESTful APIs
- Optimized database queries and improved system security and performance

Full Stack Developer | *Technolangs Solutions — Faisalabad*

Aug 2024 – Dec 2024

- Built UI-intensive projects using jQuery & Bootstrap
- Developed a MERN-based Twitch Clip Downloader with real-time data fetching

RESEARCH & AI / ML PROJECTS

Healthcare RAG for Clinical Decision Support | *MS Thesis Research — In Progress*

Designing a Retrieval-Augmented Generation (RAG) pipeline to support community health workers with Urdu-language queries grounded in national clinical guidelines, for a Pakistani healthcare context.

Satellite Imagery, LLMs & RAG for Precision Agriculture | *Research Exploration*

Investigated the integration of satellite/remote-sensing imagery with LLM-driven RAG systems for agricultural decision support. Reviewed and mapped research gaps against RS-RAG and AgroBot as reference architectures.

Agentic Multimodal RAG for Smart Agriculture | *Research Proposal*

Contributed to a multimodal RAG research proposal combining heterogeneous agricultural data sources. Identified and reviewed supporting literature, including AgroLLM, AgroOmni, and agentic-AI approaches for smart agriculture.

AI Image Authentication & Counterfeit Detection | *Client-Delivered ML Project*

Debugged a ResNet50-based CNN model for detecting fake/counterfeit brand imagery. Diagnosed a preprocessing mismatch (ImageNet normalization vs. simple pixel scaling) that was degrading model accuracy.

Computer Vision Foundations | *Applied Coursework*

Studied and applied core computer vision techniques: Canny edge detection, image gradients, and spatial filtering.

FULL-STACK DEVELOPMENT PROJECTS

- **GameXchange Platform:** Niche marketplace for game trading (Node.js, Express, React.js, MySQL).
- **Event Booking System:** End-to-end full-stack booking system alongside multiple projects spanning e-commerce tools, hospitality sites, and institutional websites.

TECHNICAL SKILLS

AI / Machine Learning: LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering, Agentic AI, Convolutional Neural Networks (ResNet)

Computer Vision: Image preprocessing, edge detection, spatial filtering, image classification pipelines

Research Exposure: GIS / Remote Sensing & satellite imagery integration with LLM/RAG systems for agriculture

Languages: C++, Java, Python, JavaScript (ES6+), PHP

Frontend: HTML, CSS, Sass, Bootstrap, jQuery, React.js

Backend: Node.js, Express.js, Laravel | **Databases:** MongoDB, MySQL

Tools: Git, GitHub, Postman, VS Code | **Teaching:** Curriculum development, session planning, mentorship, public speaking

CERTIFICATIONS & ACHIEVEMENTS

- Certified Programming Trainer – NAVTTC | Certified Web Developer – NAVTTC
- Google Analytics Certified | PM Youth Laptop Scheme Awardee (2023)
- Honoured by UCP, UAF and NIC with Guest Speaker Shields
- Event Organizer – CST Department (UAF) | Mentor – Technolangs Solutions | Python Course Moderator – iCodeGuru